DUNDALK INSTITUTE OF TECHNOLOGY

DK_ICSOP_7 - B.Sc. in Computing Systems and Operations

DK_ICCSO_8 – B.Sc. (Hons) in Computing Systems and Operations

Stage 3 - Semester 5

January 2025

COMP I8064 - Data Structures

Internal Examiner(s): Dr. Michelle Graham

External Examiner(s): xxxxxxx

Date:

Time:

Duration:

Venue:

No. Pages:

Instructions

Answer QUESTION ONE and any other TWO questions.

PLEASE NOTE: Exam Paper printed DOUBLE SIDED.

DO NOT TURN THIS PAGE UNTIL INSTRUCTED

You MUST attempt this question.

Question 1) Data Structure Fundamentals

(a) Data structures are designed to store and operate on data within a program.

(i) What is an Abstract Data Type (ADT)? What is the benefit of separating a data structure's ADT from its implementation?

(5 marks)

(ii) Formally specify the ADT for a *List* structure. Your ADT should include all fundamental operations for a *List* and explain the need for their presence.

(10 marks)

(b) There are many data structures to choose from when programming. Identify three criteria that should be considered when choosing a data structure for use in an application. Explain why these criteria are of such importance.

(9 marks)

(c) Your team has been asked to write unit tests for a singly-linked list that tracks the tail node in the list. The list is built to hold Strings, is unbounded, and duplicate values are allowed. Null values should not be allowed. You have been assigned the following method:

- public int indexOf(String s)

Outline four test cases that should be included as unit tests for the above method. For each test case, explain its necessity, what assertions should be included and why they are needed.

(16 marks)

Choose any TWO questions from questions 2, 3 and 4

Question 2) Sets

(a) What is a "Set" data structure? List and explain the fundamental operations for a *Set*.

(5 marks)

(b) Write and explain the **pseudocode** for the *add* method in a Set that has been implemented as a Binary Search Tree. The Set should contain String objects. Note: Your method should be done as pseudocode, not actual code. Pseudocode and explanation are worth equal marks.

(12 marks)

(c) Is implementing a Set as a Binary Search Tree an efficient approach? Explain your answer, supported with reference to how the add, remove and check functionality of a Set would be provided in a Binary Search Tree.

(8 marks)

(d) Outline an alternative approach to implementing a Set. Is this approach an improvement on implementing it as a Binary Search Tree? Justify your answer.

(5 marks)

Question 3) Maps

(a) A Hash Map is a commonly used data structure that allows for fast data retrieval and storage.

(i) Briefly explain a Hash Map. How is it different from a standard array?

(5 marks)

(ii) Describe the **hashing process** used in a Hash Map to store and retrieve values. Include an explanation of how a **hash function** works.

(5 marks)

(b) What is meant by the term *collision* in relation to a Hash Map? Explain the main causes of collisions in Hash Maps. What is the impact of *collisions* in a Hash Map without collision resolution?

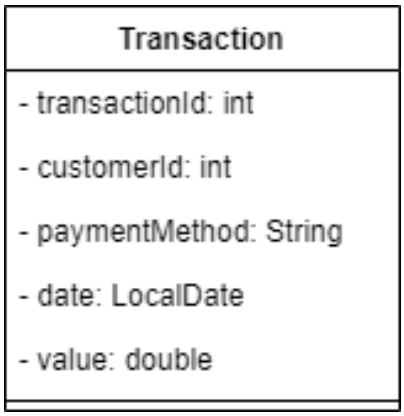
(6 marks)

(c) Compare and contrast two approaches you have studied to collision resolution when inserting into Hash Maps. Your answer should include a discussion of the time and space complexities of each approach.

(14 marks)

Question 4) Analysis of Data Structures

(a) Your team has been assigned the task of building a Point of Sale system. You have been asked to research suitable data structures for the storage of Transaction information in memory (a database approach has been deemed unsuitable for this situation). A Transaction is modelled as:



(i) Compare and contrast the suitability of the following structures as transaction storage for this system:

- Collision-resistant HashMap
- Link-based List (maintaining head and tail nodes)
- Link-based Queue

Your answer should include a discussion of time and space complexities. It should consider:

(i) Insertion.
(ii) Deletion.
(iii) Retrieval.

(27 marks)

(ii) Recommend, with reference to your answer to 4a) which structure should be employed if the system is to be deployed in a low memory environment where Transactions are frequently required to be retrieved in order of transaction date.

(3 marks)